



RESEARCH METHODOLOGY & DATA DESCRIPTION

Computer Vision-Based Sri Lankan Food Recognition and Calorie Estimation System

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1 DATASET DESCRIPTION

1.1 Source and Collection Procedure

The main dataset for this research will be a locally collected image dataset of Sri Lankan meals. The dataset will be created specifically for this project because existing public datasets do not fully support the requirements of this study, especially mixed meal recognition and portion size estimation.

The data collection will be carried out using two main approaches.

1. Field Data Collection

We will collect food images by taking photographs of Sri Lankan meals from different real-world environments, including university canteens, home kitchens, and local food outlets. The images will be captured using smartphone cameras under normal lighting conditions to represent realistic meal situations.

Each meal image will be taken from a top-view angle to maintain consistency. A standard reference object, such as a coin or ArUco marker, will be placed next to the plate during image capture. This reference object will help estimate the portion size of food items based on pixel measurements instead of making assumptions about food quantity.



Figure 1: Food Image Capture with Reference Object

The main collection locations will include Horizon Campus canteen, local rice and curry restaurants around Malabe town, food outlets near Kaduwela area, and our home kitchens. In addition to images collected by both of us, our friends and family members may also contribute food images by following the same image collection guidelines to increase the diversity of the dataset [1].

2. Use of Public Reference Dataset

The publicly available Roboflow "SriLankanFoods" dataset will be used as a supplementary dataset for validation and comparison purposes. It will not be considered the main training dataset because its available food categories and image quantity are not sufficient for the requirements of this research, particularly for mixed Sri Lankan meals and portion size estimation.

The public dataset will help evaluate whether the locally collected images have similar characteristics compared with existing Sri Lankan food image datasets. It will also support testing the general performance of the developed model.

Therefore, the primary dataset used in this study will be the locally collected Sri Lankan meal image dataset, while the Roboflow dataset will only be used as an additional reference source.

1.2 Size and Class Distribution

The target dataset size is approximately 3,000 - 4,000 images covering around 18 - 20 Sri Lankan food classes. The selected classes will include common Sri Lankan meals such as rice and curry, kottu roti, string hoppers, hoppers, pittu, lamprais, sambols, and different vegetable, fish, and meat curries.

Since a typical Sri Lankan meal plate contains several food items together, each image will be labelled with multiple food categories. For example, one image can contain labels such as rice, dhal curry, and pol sambol at the same time. This allows the model to recognize multiple food items from a single meal image.

The number of images may not be equal for all food classes because some meals are more common than others. To handle this issue, the training process will give more importance to less common food classes, and the dataset will be divided carefully during testing to include both common and rare food categories [2], [3].

After completing data collection, the final number of images and the distribution of images across each food class will be updated in the report with a simple class distribution chart.

1.3 Ethical Considerations

This dataset contains only food images and does not collect personal information about people. Therefore, privacy risks are limited. However, some ethical steps will be followed during data collection.

Images taken in shared places such as food outlets or canteens will focus only on the food plate. If any people appear in the background, the image will be cropped or removed. Permission will be obtained from relevant staff or food owners before taking photographs, and no images will be collected without consent.

The dataset will store only the meal image, food class labels, and estimated calorie values. No personal details about customers or food providers will be collected.

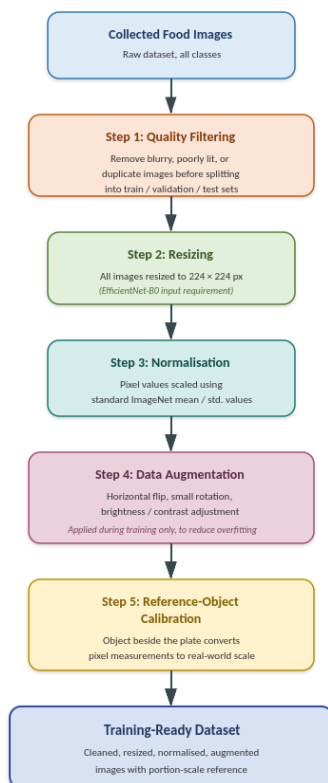
1.4 Annotation Process

The food images will be labelled by the two researchers based on their knowledge of Sri Lankan dishes. Each image will be reviewed and labelled separately by both annotators. If there are any differences between labels, they will discuss and agree on the final label. The agreement between annotators will also be measured to ensure the reliability of the dataset.

Calorie estimation will be performed using a recognized food nutrition reference and standard portion sizes for each dish. Since food portions can vary between meals, the calorie values will be estimated using the portion size calculated from the image reference object. These values will be considered approximate rather than exact measurements, and this limitation will be included in the evaluation [4].

1.5 Preprocessing

Before training the AI model, all collected images will go through preprocessing steps to improve consistency and quality [5].



- Low-quality images such as blurry, poorly lit, or duplicate images will be removed before dividing the dataset into training, validation, and testing sets.
- All images will be resized to 224×224 pixels to match the input requirement of the EfficientNet-B0 model.
- Pixel values will be normalized using ImageNet standard values to support the pretrained model.
- Data augmentation techniques such as horizontal flipping, small rotations, and brightness/contrast adjustments will be applied during training to improve model performance and reduce overfitting.
- The reference object placed beside the plate will be used to estimate food portion size by converting image pixel measurements into approximate real-world measurements.

Figure 2: Image Preprocessing Pipeline

2 MODEL ARCHITECTURE

The proposed system uses a two-headed deep learning model with a shared **EfficientNet-B0** backbone. One branch is responsible for identifying food items, while the second branch estimates the portion size of each food region. The final calorie value is calculated by combining the predicted food classes, estimated portion sizes, and nutrition information from a reference database.

The model uses EfficientNet-B0 pretrained on ImageNet as the main feature extraction network. It is selected because it provides good accuracy with fewer parameters compared with larger models, which is suitable for a medium-sized dataset and helps reduce overfitting.

The classification branch uses a multi-label approach with sigmoid activation because a single Sri Lankan meal image can contain multiple food items at the same time. This allows the model to identify several dishes from one plate. The portion estimation branch uses shared image features to identify food regions and estimate their sizes. The reference object placed during image collection is used to convert pixel measurements into approximate portion sizes. The calorie estimation step combines the detected food items, estimated portions, and nutrition reference values to calculate the final calorie value. This rule-based approach improves interpretability and avoids the need for a large amount of accurate calorie-labelled data [6], [7].

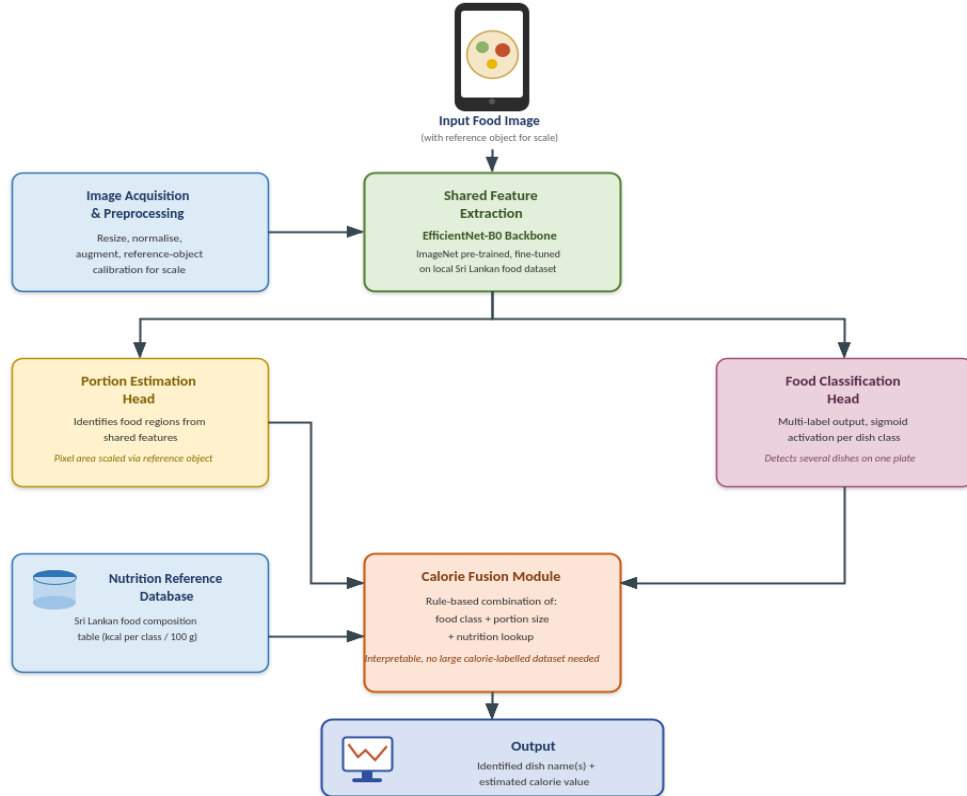


Figure 3: Proposed EfficientNet-B0 Architecture for Sri Lankan Food Classification and Portion-aware Calorie Estimation

3 BASELINE MODEL

The baseline model for this research is **ResNet-50**, pretrained on ImageNet and fine-tuned using the same local Sri Lankan food image dataset. Unlike the proposed model, the baseline uses only a single multi-label classification head to identify food items and does not estimate portion sizes.

For calorie estimation, the baseline uses a fixed average calorie value for each detected food class without considering the actual portion size of the meal.

ResNet-50 was selected as the baseline because it is a well-known and widely used deep learning model for image classification. Both the baseline model and the proposed EfficientNet-B0 model will be trained and evaluated using the same dataset and the same training, validation, and testing procedure. This ensures that the comparison is fair and that any performance differences are due to the model architecture rather than the dataset.

By comparing these two models, this research will evaluate whether the proposed multi-head architecture with portion size estimation provides better food recognition and calorie estimation than a standard transfer learning approach [8].

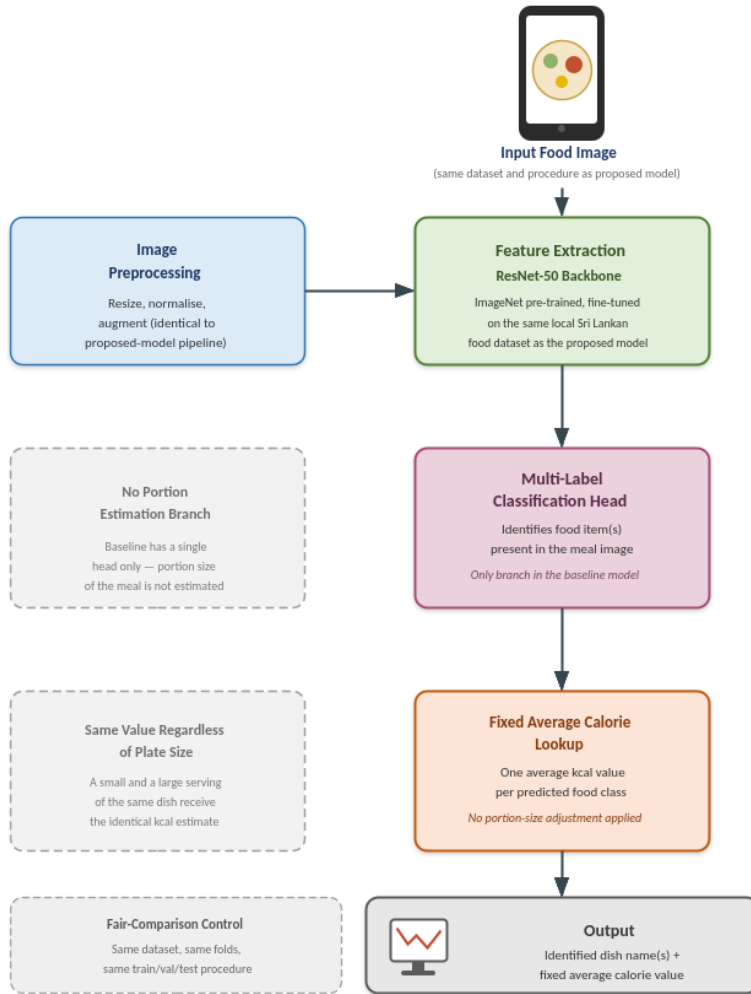


Figure 4: Baseline ResNet-50 Model Architecture

4 EVALUATION PLAN

4.1 Evaluation Metrics

The performance of the proposed model will be evaluated using both classification and regression metrics to assess its effectiveness in identifying Sri Lankan food dishes and estimating their calorie content.

For dish classification, the **Macro-averaged F1-score** will be used as the primary evaluation metric. Since the dataset contains an imbalanced distribution of food categories, relying solely on accuracy could produce misleading results by favoring the more frequently occurring classes. The Macro-F1 score addresses this issue by giving equal importance to every class, providing a more balanced measure of classification performance [9].

As a secondary metric, **AUC-ROC (One-vs-Rest, Macro-Averaged)** will be used to evaluate the model's ability to distinguish between different dish categories. This metric complements the Macro-F1 score by measuring the model's discriminative capability across all classes, regardless of the decision threshold [10].

For calorie estimation, **Mean Absolute Error (MAE)** and **Mean Absolute Percentage Error (MAPE)** will be used. MAE measures the average difference between the predicted and actual calorie values in kilocalories, making the results easy to interpret. MAPE expresses the prediction error as a percentage of the true calorie value, allowing the estimation accuracy to be evaluated across meals with different portion sizes [11], [12].

Task	Metric	Purpose
Food Classification	Macro-F1 Score	Measures balanced classification performance across all food classes
Food Classification	AUC-ROC	Evaluates class discrimination ability
Calorie Estimation	MAE	Measures average calorie prediction error in kcal
Calorie Estimation	MAPE	Measures relative prediction error percentage

Table 1: Evaluation Metrics Used in the Study

4.2 Validation Strategy

A **stratified 5-fold cross-validation** approach will be used to evaluate the proposed model. The dataset will be divided into five folds while maintaining a similar distribution of the primary dish class in each fold. During each iteration, four folds will be used for training and the remaining fold will be used for validation. This process will be repeated five times so that every fold serves as the validation set once [13].

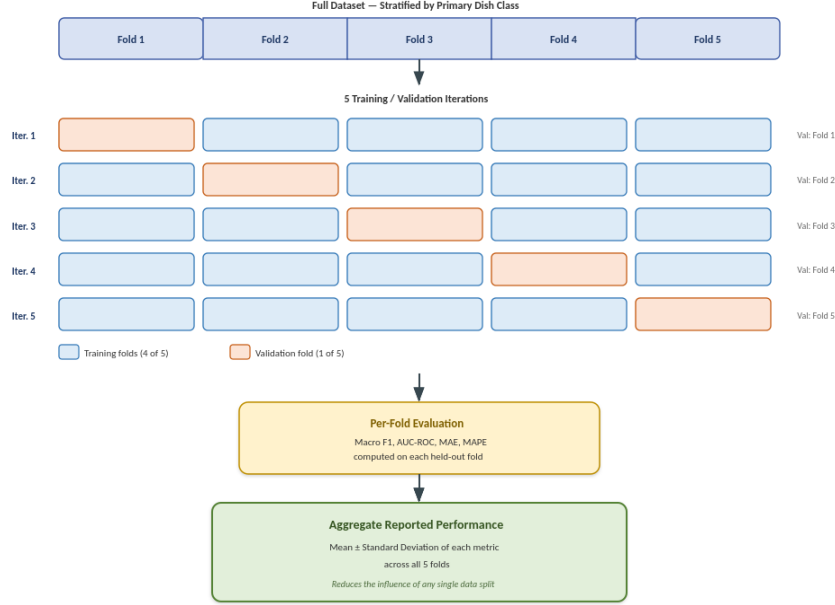


Figure 5: Stratified 5-Fold Cross Validation Process

This validation strategy is selected because the dataset is relatively small and contains an uneven distribution of food classes. Compared with a single train-test split, stratified 5-fold cross-validation provides a more reliable estimate of model performance by reducing the influence of a particular data split. The final performance will be reported as the mean and standard deviation of the evaluation metrics across all five folds.

4.3 Statistical Testing

The performance of the proposed **EfficientNet-B3** model will be compared with the **ResNet-50** baseline. The comparison will focus on the Macro-F1 score for dish classification and the MAE for calorie estimation, using the results obtained from the same five cross-validation folds.

Before selecting the statistical test, the normality of the paired differences between the two models will be assessed using the **Shapiro–Wilk test**. If the differences are approximately normally distributed, a **paired t-test** will be performed. Otherwise, the **Wilcoxon signed-rank test** will be used as a non-parametric alternative that does not require the normality assumption. In addition to reporting statistical significance, the mean and standard deviation of the evaluation metrics will also be presented to provide a clearer understanding of the performance and consistency of the proposed model [14], [15].

5 GITHUB REPOSITORY

The public GitHub repository for this research project is available at:

GitHub Repo Link: [GitHub Repository](#)

The repository contains the initial implementation, documentation, and project resources developed for this milestone. It provides the required project structure, preprocessing implementation, and supporting files to ensure reproducibility and demonstrate the progress of the research development.

The current repository includes the following components:

- **README.md** - Contains the project title, research background, current development status, project structure, and instructions for running the preprocessing scripts.
- **data/** - Contains the dataset directory structure for storing raw and processed Sri Lankan food images. Due to the large size of image datasets, raw and processed image files will not be directly committed to the repository and are excluded using the .gitignore file.
- **scripts/preprocessing/** - Contains the image preprocessing scripts required for preparing the dataset. These scripts include image resizing, normalization, data augmentation, and reference-object calibration for portion size estimation.
- **notebooks/** - Contains exploratory notebooks used for dataset analysis, preprocessing experiments, and initial investigation of the collected food images.
- **docs/** - Contains research documentation files, including the Milestone 2 report and the source file of the proposed system architecture diagram.
- **requirements.txt** - Contains the required Python libraries and dependencies needed to run the preprocessing pipeline and support future model development, including deep learning frameworks and computer vision libraries.

The repository development is maintained through incremental commits to demonstrate continuous progress throughout the research process.

6 REVISED BIBLIOGRAPHY

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7 ACADEMIC INTEGRITY DECLARATION

IT41043 - Intelligent Systems | Assignment 1: Research Proposal

I declare that this report is our own original work. I confirm that:

1. I have not used any large language model (LLM), generative AI tool, or AI writing assistant to draft, outline, paraphrase, or edit any portion of this document.
2. I have personally read every source included in my annotated bibliography.
3. All sources are correctly attributed. I have not fabricated, misrepresented, or plagiarised any reference.
4. I understand that submitting work that is not my own constitutes academic misconduct and may result in a zero mark and disciplinary action under the Horizon Campus Academic Integrity Policy.

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